AI Water Allocation

Project Documentation

1. Project Planning & Management

1.1 Project Proposal

1.1.1 Project Overview

AI Water Allocation is a proposed intelligent agricultural decision-support system designed to help farmers use irrigation water more efficiently. The system will combine agricultural data, weather conditions, soil information, and crop-condition analysis to estimate irrigation water requirements and recommend suitable water allocation decisions.

The proposed system will integrate machine learning for water requirement prediction, computer vision for analyzing crop images, and optimization algorithms for allocating limited water resources. A user-friendly dashboard will present the resulting recommendations, estimated water requirements, and relevant information to support decision-making.

The project aims to demonstrate how artificial intelligence and data-driven optimization can contribute to more efficient and sustainable agricultural water management.

1.1.2 Problem Statement

Agricultural irrigation decisions are affected by several factors, including crop type, growth stage, soil moisture, temperature, humidity, and rainfall. Using fixed irrigation schedules or relying solely on manual judgment may lead to inefficient water use, particularly when water resources are limited.

The project addresses the need for a data-driven system that estimates crop water requirements and recommends irrigation allocations based on available agricultural and environmental information.

1.1.3 Project Objectives

The main objectives are:

1. Estimate irrigation water requirements using relevant crop, soil, and weather data.
2. Analyze crop images to identify visible crop conditions that may indicate water stress.
3. Apply machine learning techniques to support irrigation requirement prediction.

4. Develop an optimization component that recommends water allocation while considering resource constraints.
5. Develop an interactive dashboard to display predictions, recommendations, and relevant indicators.
6. Evaluate the system using appropriate prediction metrics and water-allocation scenarios.
7. Promote more informed and efficient agricultural water management.

1.1.4 Project Scope

Included in the project

- Collecting and preparing suitable agricultural image and tabular datasets.
- Processing crop, soil, and weather information.
- Developing a machine learning model for estimating irrigation requirements.
- Developing a computer vision component for crop-condition analysis.
- Formulating an optimization problem for allocating limited irrigation water.
- Integrating the components into a dashboard.
- Evaluating model performance and comparing allocation scenarios.

Outside the initial scope

- Manufacturing and deploying physical irrigation hardware.
- Guaranteeing fully autonomous irrigation in real farms.
- Providing professional agronomic diagnoses based solely on images.
- Guaranteeing water savings or improved crop yields without field validation.

The initial implementation will focus on a software-based prototype using available datasets. The final scope may be refined according to data availability, project duration, and technical feasibility.

1.2 Project Plan

1.2.1 Phases and Deliverables

The project will follow an iterative development approach consisting of planning, research, requirements analysis, data preparation, model development, optimization, integration, testing, and documentation.

Phase	Main activities	Expected deliverable
Planning	Define objectives, scope, roles, risks, and timeline	Project planning document
Literature review	Research existing irrigation and agricultural AI solutions	Literature review
Requirements gathering	Identify stakeholders, features, and constraints	Requirements specification
System analysis and design	Define architecture, workflows, and data flow	Design documentation
Data preparation	Collect, clean, and prepare datasets	Prepared datasets
Model development	Develop and evaluate prediction and vision components	Trained and evaluated models
Optimization	Formulate and test water-allocation scenarios	Optimization component
Integration	Connect components through an interactive dashboard	Integrated prototype
Testing and evaluation	Test functionality and evaluate performance	Test results and evaluation report
Final delivery	Finalize code, documentation, and presentation	GitHub repository and final presentation

1.2.2 Timeline (Gantt Chart)

The chart below shows the proposed schedule. The documentation deadlines follow the official DEPI schedule; the start and end weeks of the internal implementation phases are proposed and may be adjusted by the team.

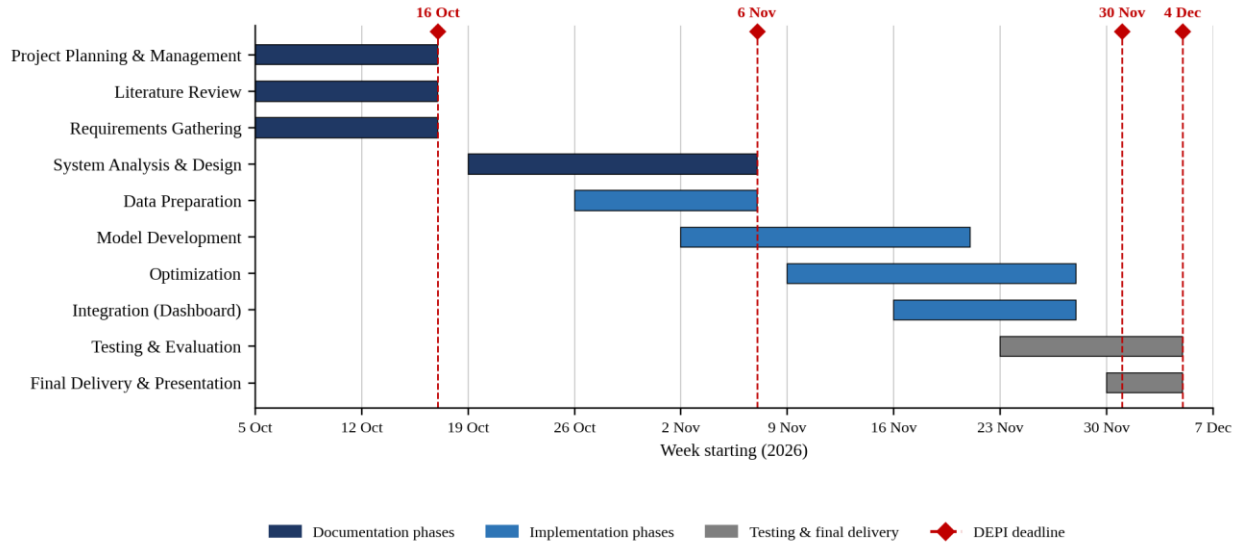


Figure 1: Proposed project timeline (Gantt chart)

1.2.3 Milestones

Milestone dates follow the official DEPI project schedule.

Milestone	Deliverables	Due date
Project Planning & Management	Project proposal, project plan, task assignment and roles, risk assessment, KPIs	16 Oct 2026
Literature Review	Review of existing solutions, feedback and evaluation, suggested improvements, grading criteria	16 Oct 2026
Requirements Gathering	Stakeholder analysis, user stories and use cases, functional and non-functional requirements	16 Oct 2026
System Analysis & Design	Design documentation (architecture, workflows, and data flow)	6 Nov 2026
Implementation (Source Code & Execution)	Source code, prepared datasets, trained models, optimization component, integrated dashboard prototype	30 Nov 2026
Final Presentation & Testing & Reports	Test results and evaluation report, GitHub repository, final presentation	4 Dec 2026

1.2.4 Resource Allocation

Human resources

Team member	Main area of responsibility	Primary project phases
Ahmed Hamdy	Data collection, cleaning, and preparation	Data preparation
Saeed	Computer vision and crop-image analysis	Model development
Seif	Machine learning for irrigation water requirement prediction	Model development
Ahmed Youssef	Irrigation optimization and allocation constraints	Optimization
Ziad Desoky	Dashboard development and economic analysis	Integration
All team members	Planning, literature review, requirements, design, integration, testing, and final documentation	All phases

Data, tools, and infrastructure

Resource	Purpose	Notes
Agricultural datasets	Labeled crop images and tabular crop, field, soil, and weather data	Public or otherwise authorized sources; licenses to be documented
Reference irrigation measurements	Train and evaluate the prediction models	Where available
Computing resources	Train, evaluate, and run the models and dashboard	Availability to be confirmed by the team
GitHub repository	Host the source code and all project documents	Required for all documents

1.3 Task Assignment & Roles

Team member	Proposed responsibility
Ahmed Hamdy	Data collection, cleaning, and preparation
Saeed	Computer vision and crop-image analysis
Seif	Machine learning for irrigation water requirement prediction
Ahmed Youssef	Irrigation optimization and allocation constraints
Ziad Desoky	Dashboard development and economic analysis

Responsibilities will be coordinated across the team, with integration and testing performed collaboratively. These assignments should be confirmed by the team and communicated to the Technical Instructor.

1.4 Risk Assessment & Mitigation Plan

Risk	Potential impact	Mitigation
Insufficient or unsuitable datasets	Poor model performance	Use reputable public datasets, inspect data quality, and reduce scope if necessary
Missing irrigation requirement labels	Difficulty evaluating predictions	Investigate reliable reference calculations or suitable labeled datasets
Inconsistent data formats	Integration problems	Standardize units, column names, and preprocessing procedures
Computer vision cannot reliably identify water stress	Incorrect crop-condition interpretation	Use appropriate labels and clearly state model limitations
Prediction model performs poorly	Unreliable recommendations	Establish a baseline, compare models, and evaluate using appropriate metrics
Optimization produces infeasible allocations	Invalid recommendations	Define water constraints explicitly and test edge cases
Integration delays	Incomplete prototype	Develop components independently and integrate incrementally
Team delays or uneven workload	Missed milestones	Assign clear ownership, track progress, and review tasks regularly

1.5 Key Performance Indicators (KPIs)

The following indicators will be used to evaluate the project:

KPI	Evaluation method
Water requirement prediction accuracy	MAE and RMSE against reference values
Model explanatory performance	R^2 where appropriate
Crop-condition classification performance	Precision, recall, F1-score, and confusion matrix
Water allocation feasibility	Verify that recommendations satisfy defined resource constraints
Potential water savings	Compare simulated allocation with a clearly defined baseline
Dashboard functionality	Test required features and user workflows

KPI	Evaluation method
System responsiveness	Measure prediction and recommendation generation time

Numerical performance targets will be defined after examining the datasets and establishing baseline results. Simulated water savings will be reported as estimates rather than proven real-world outcomes.

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2. Literature Review

2.1 Review of Existing Solutions and Research

2.1.1 FAO Crop Evapotranspiration Methodology

Allen et al. (1998), in the Food and Agriculture Organization's Irrigation and Drainage Paper 56, describe a widely used methodology for estimating crop evapotranspiration. The approach estimates crop evapotranspiration by multiplying reference evapotranspiration (ETo) by a crop coefficient (Kc), which varies according to crop type and growth stage.

The methodology provides a scientific foundation for estimating agricultural water requirements using meteorological data and crop characteristics. However, applying the methodology requires appropriate input data and does not, by itself, provide the complete integrated dashboard and constrained water-allocation workflow proposed in AI Water Allocation.

2.1.2 IoT-Based Smart Irrigation Management

Goap et al. (2018) proposed an IoT-based smart irrigation management system using machine learning and open-source technologies. Their approach combines soil and environmental sensor readings with weather forecasts to support soil-moisture prediction and irrigation scheduling.

The system demonstrates the value of integrating environmental observations with predictive algorithms. A consideration for implementation is the dependence on suitable sensor readings and weather data. AI Water Allocation will initially emphasize a software-based prototype using available datasets, rather than requiring the deployment of physical sensors.

2.1.3 AI-Based Estimation of Crop Water Use and Stress

A 2023 study published in Smart Agricultural Technology evaluated artificial intelligence algorithms for estimating crop evapotranspiration and crop water stress indicators. The study used combinations of sensor and environmental data, including weather measurements, soil moisture, canopy temperature, and vegetation indices.

This research supports the use of data-driven models to assist irrigation management. However, model performance depends on the availability and quality of input data and appropriate reference measurements. The proposed project will investigate how prediction, crop-image analysis, and constrained water allocation can be combined into a single decision-support workflow.

2.2 Comparison of Existing Approaches

The reviewed approaches address different parts of the irrigation decision-making process.

- The FAO methodology provides a scientific basis for estimating crop water demand.
- IoT-based smart irrigation systems combine environmental monitoring with predictive irrigation scheduling.
- AI-based crop water-use research investigates how machine learning can estimate evapotranspiration and water-stress indicators.

These approaches provide useful foundations for the proposed project. They also highlight the importance of reliable environmental data, appropriate model evaluation, and irrigation recommendations that reflect crop and field conditions.

2.3 Research Gap and Proposed Contribution

AI Water Allocation proposes an integrated software-based decision-support workflow that combines agricultural data processing, crop-condition image analysis, irrigation requirement prediction, and water-allocation optimization.

Rather than treating prediction as the final output, the system aims to use predicted irrigation requirements as inputs to an optimization component that considers available water and defined allocation constraints. An interactive dashboard will present the resulting recommendations and relevant performance indicators.

The project's contribution will be evaluated through implementation and testing. Any claimed improvement in prediction accuracy or water savings will require comparison against suitable baselines and supporting experimental results.

2.4 Feedback & Evaluation

This section is reserved for the lecturer's assessment of the project and is to be completed after the lecturer reviews the submitted work.

Aspect	**Lecturer's comments**
Overall assessment	

Aspect	Lecturer's comments
Strengths	
Areas needing attention	
Reviewer name and date	

2.5 Suggested Improvements

Based on the limitations and research gaps identified in this document, the following areas could enhance the project. Further suggestions may be added following the lecturer's feedback.

- **Field validation** – Validate predictions and simulated water savings using real field measurements, since savings and yield benefits cannot be guaranteed without field validation.
- **Sensor data integration** – Extend the software prototype to use soil and environmental sensor readings when available, as demonstrated by IoT-based irrigation research.
- **FAO-based reference calculations** – Use the ETo and Kc methodology to generate or verify reference irrigation requirements where labeled data is missing.
- **Broader image coverage** – Expand the crop and crop-condition categories supported by the computer vision component and improve label quality.
- **Real-world deployment** – Explore connecting recommendations to irrigation equipment, which is currently outside the initial scope.

2.6 Final Grading Criteria

The breakdown of marks across documentation, implementation, testing, and presentation is to be completed by the lecturer.

Category	Covers	Marks
Documentation	Project planning, literature review, requirements, system analysis and design, user manual, and technical documentation	
Implementation	Source code quality, version control, and deployment and execution	
Testing	Test cases and test plan, automated tests, and bug reports	
Presentation	Project presentation, project infographic, and optional video demonstration	

Category	Covers	Marks
Total		

3. Requirements Gathering

3.1 Stakeholder Analysis

The primary stakeholders of AI Water Allocation are:

Stakeholder	Needs and expectations
Farmers	Understand estimated irrigation requirements and receive clear recommendations
Agricultural engineers and specialists	Review crop, soil, and weather information and interpret recommendations
Farm managers	Allocate limited water resources across fields according to defined priorities and constraints
Project development team	Access suitable datasets, develop the models, integrate the system, and evaluate its performance
Technical Instructor and evaluators	Review the project methodology, implementation, documentation, and evaluation results

The initial prototype will focus on supporting agricultural decision-making rather than replacing professional agricultural expertise.

3.2 User Stories & Use Cases

3.2.1 User Stories

The user stories below describe how each stakeholder is expected to interact with the system and are linked to the functional requirements in Section 3.3.

ID	As a…	I want to…	So that…	Related FR
US-01	Farmer	enter my crop, field, soil, and weather information	I receive an estimate of how much irrigation water my crops need	FR-01, FR-04
US-02	Farmer	be told when my inputs are missing or invalid	I can correct them before relying on a result	FR-02

ID	As a…	I want to…	So that…	Related FR
US-03	Agricultural specialist	upload a crop image	I can see the estimated visible crop condition	FR-03
US-04	Farm manager	specify the available water supply and allocation constraints	recommendations reflect the real limits on my water	FR-05
US-05	Farm manager	receive a recommended water allocation across fields	I can distribute limited water within my constraints	FR-06
US-06	Farmer	view predictions and recommended allocations on a dashboard	I can understand the results clearly	FR-07
US-07	Farm manager	compare allocation scenarios against a baseline	I can judge the potential benefit of a recommendation	FR-08
US-08	Agricultural engineer	see the limitations or missing information behind a recommendation	I can interpret it with appropriate caution	FR-09
US-09	Project team member	evaluate the prediction models and optimization component with appropriate metrics	I can verify system quality	FR-10
US-10	Technical Instructor	review the evaluation results and documentation	I can assess the project's methodology and outcomes	FR-10

3.2.2 Use Cases

The primary actors are the farmer, the agricultural engineer or specialist, and the farm manager. The development team and evaluators interact with the system through model evaluation.

ID	Use case	Primary actor	Description
UC-01	Provide field and weather data	Farmer, farm manager	Enter crop, field, soil, and weather information; the system validates the inputs
UC-02	Analyze crop image	Farmer, agricultural specialist	Upload a supported crop image to obtain the estimated crop condition
UC-03	Estimate irrigation requirement	Farmer, agricultural engineer	Obtain the estimated irrigation water requirement for a field or scenario
UC-04	Generate water allocation	Farm manager	Specify available water and constraints to receive a recommended allocation

ID	Use case	Primary actor	Description
UC-05	View dashboard results	All users	Review predictions, allocations, input information, and stated limitations
UC-06	Compare allocation scenarios	Farm manager	Compare scenarios against a defined baseline using relevant indicators
UC-07	Evaluate models	Project team member, evaluators	Review model and optimization performance using metrics and test cases

UC-03: Estimate irrigation requirement

Item	Description
Actor	Farmer or agricultural engineer
Precondition	The prediction approach has been developed and the user has crop, field, soil, and weather information available.
Main flow	1. The user enters the crop, field, soil, and weather information. 2. The system validates the inputs and identifies missing or invalid values. 3. The system estimates the irrigation water requirement. 4. The dashboard displays the estimate with explicit units and relevant limitations.
Alternative flow	If inputs are missing or invalid at step 2, the system identifies them and asks the user to correct them before continuing.
Postcondition	The user can view the estimated requirement and any stated limitations.
Related requirements	FR-01, FR-02, FR-04, FR-07, FR-09

UC-04: Generate water allocation

Item	Description
Actor	Farm manager
Precondition	Irrigation requirements have been estimated for the supported fields or scenarios.
Main flow	1. The user specifies the available water supply and allocation constraints. 2. The system uses the estimated requirements as inputs to the optimization component. 3. The system generates a water allocation that satisfies the constraints. 4. The dashboard displays the recommended allocation and comparison indicators.

Item	Description
Alternative flow	If no feasible allocation exists for the specified constraints, the system informs the user so the constraints can be reviewed.
Postcondition	The user can review the recommended allocation and compare it with a defined baseline.
Related requirements	FR-05, FR-06, FR-07, FR-08, FR-09, NFR-05

3.3 Functional Requirements

Functional requirements describe the actions and features that the system must provide.

ID	Requirement	Priority
FR-01	The system shall accept relevant crop, field, soil, and weather information.	High
FR-02	The system shall validate inputs and identify missing or invalid values.	High
FR-03	The system shall process supported crop images to estimate visible crop condition or classify supported categories.	High
FR-04	The system shall estimate irrigation water requirements using the available input features and the developed prediction approach.	High
FR-05	The system shall allow the user to specify the available water supply and relevant allocation constraints.	High
FR-06	The system shall generate water-allocation recommendations that satisfy the defined constraints whenever a feasible solution exists.	High
FR-07	The system shall display predictions, recommended allocations, and relevant input information through a dashboard.	High
FR-08	The system shall provide relevant indicators for comparing allocation scenarios against a defined baseline.	Medium
FR-09	The system shall communicate limitations or missing information that may affect recommendations.	High
FR-10	The system shall support evaluation of the prediction models and optimization component using appropriate metrics and test cases.	High

3.4 Non-Functional Requirements

Non-functional requirements describe the expected quality of the system.

ID	Requirement	Verification method
NFR-01	Performance: The system should generate results within a practical response time for an interactive prototype.	Measure execution time on representative inputs
NFR-02	Reliability: The system should handle invalid inputs and missing values without unexpected crashes.	Test invalid, incomplete, and boundary-case inputs
NFR-03	Usability: The dashboard should present information using clear labels, units, and understandable recommendations.	Conduct basic usability checks
NFR-04	Accuracy: Prediction models should be evaluated using suitable metrics and held-out test data.	Report MAE, RMSE, and other appropriate metrics
NFR-05	Feasibility: Recommended allocations must respect the specified water-supply and allocation constraints.	Test optimization constraints across multiple scenarios
NFR-06	Maintainability: Data preparation, prediction, optimization, and dashboard components should be organized into understandable modules.	Review project structure and code organization
NFR-07	Security: The application should validate user inputs and avoid exposing credentials or sensitive information.	Review input handling and repository contents
NFR-08	Reproducibility: Experiments should document datasets, preprocessing steps, model configurations, and evaluation methods.	Review experiment documentation

Specific response-time and accuracy thresholds will be established after initial benchmarking and consideration of the available hardware and datasets.

3.5 Data Requirements

The system will require suitable data for each of its major components.

Data category	Required information	Purpose
Crop images	Labeled images of supported crops and relevant crop conditions	Computer vision and crop-condition analysis
Crop information	Crop type and, where available, growth stage	Estimate crop-specific water requirements
Field information	Field area and relevant field characteristics	Calculate water volumes and allocation needs

Data category	Required information	Purpose
Soil information	Soil type and soil-moisture measurements, where available	Characterize field water conditions
Weather information	Temperature, humidity, rainfall, and other relevant meteorological variables	Estimate environmental water demand
Reference targets	Reliable irrigation requirement or water-use measurements, where available	Train or evaluate prediction models
Water constraints	Available water supply, allocation limits, and any defined priorities	Optimize allocation across fields

Data will be checked for missing values, inconsistent units, duplicates, and unsuitable records. Image datasets will be checked for label quality and relevance to the intended task. Data sources, licenses, and usage restrictions will be documented.

3.6 System Inputs and Outputs

Inputs

- Crop type and growth stage, when available.
- Field area and relevant soil characteristics.
- Weather observations or forecasts.
- Soil-moisture measurements, when available.
- Supported crop images.
- Available water supply and allocation constraints.

Outputs

- Estimated irrigation requirement for each supported field or scenario.
- Crop-condition classification or indicators, where supported by the image model.
- Recommended water allocation under the specified constraints.
- Summary of allocation decisions and relevant indicators.
- Model evaluation results and comparisons against selected baselines.

Water quantities will be displayed with explicit units. Recommendations will be accompanied by appropriate limitations, especially when inputs are incomplete or the model has not been validated under real-world field conditions.

3.7 Assumptions and Constraints

The initial project assumes that suitable public or otherwise authorized datasets can be obtained and that sufficient information is available to demonstrate the major components.

The project is constrained by the availability and quality of labeled images, weather and soil data, reference irrigation measurements, development time, and computing resources.

The prototype will initially provide software-based recommendations. Physical sensor deployment, automated control of irrigation equipment, and real-world agronomic validation are not guaranteed deliverables unless the team explicitly adds them to the approved scope.